JODDB
المركز الأردني للتصميم والتطوير
JORDAN DESIGN & DEVELOPMENT BUREAU

المركز الأردني
للتصميم والتطوير

Number : JODDB/Sales/2022/5217
Date : 21 September 2022

Special And Safety Work Nisreen Factory/KSA
Subject: <u>Test Report</u>

Ref: Our letter number JODDB/Sales/2022/4820 Dated 1st September 2022

Dear Sir,

1. Kindly find attached Summary Test report no. JODDB/Test/BTF/STR/1086 which represents the result for the test in the reference above.

2. We need to receive your kind feedback regarding attached test report by filling the attached Customer feedback Survey.

3. Thank you for your valued business and look forward hearing from you soon.

Sincerely Yours,

On behalf of CEO Jordan Design & Development Bureau
Head of Sales & Business Strategy Department
Eng. Rateb A. Abu Al Ragheb

Cc:
- Head of Test and Evaluation Department
- CEO office manager / FYI

Test & Evaluation Department

Ballistic Testing Facility

Summary Test Results

Bullet Attack Resistance Test

Customer Requirements
Three Strikes 308 Win "P 80" from 10m (Each Sample)

Combined 14+6 mm Mild Steel Plates
46mm air Gape
Combined 14+6mm Mild Steel Plates

Combined 14+8 mm Mild Steel Plates
46mm air Gape
Combined 14+8 mm Mild Steel Plates

Combined 14mm Mild Steel Plate + 6.25 (Hardox 500) mm
46mm air Gape
Combined 14+6 mm Mild Steel Plates

Special And Safety Work Nisreen Factory / KSA

JODDB/TEST/BTF/STR/1086

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Approval

Summary Test Results

Bullet Attack Resistance Test

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Combined 14mm Mild Steel Plate + 6.25 (Hardox 500) mm
46mm air Gape
Combined 14+6 mm Mild Steel Plates

Special And Safety Work Nisreen Factory / KSA

Compiled by
Head of Weapons, Ammunition & Armor Testing
Eng. Issa Abed AL Rahman Rawashdeh.

Approved by
Head of Test & Evaluation Department.
Dr. Eng. Riyad Ali Ratrout

Signature 20 September 2022
Date

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1. Test Data

Requester	JODDB/ Sales
Contractor	Special And Safety Work Nisreen Factory HRMM+WHV, As Samh Ibn Malik, Ad Difa, Riyadh 14315, KSA
References	• Test and Evaluation Request No. 1086 dated 30/08/2022 • Test Proposal No .JODDB/TEST/BTF/TP/1086 Dated 31/08/2022 **(See Appendix A)**
Test Samples	• **Sample No.1:** Combined 14+6 mm Mild steel plates with 46mm air Gape Combined 14+6mm Mild steel plate Size 500x500mm. • **Sample No.2:** Combined 14+8 mm Mild steel plates with 46mm air Gape Combined 14+8mm Mild steel plates Size 500x500mm • **Sample No.3:** Combined 14mm Mild steel plate + 6.25 (Hardox 500) mm with 46mm air Gape Combined 14+6 mm Mild steel plates/ Size 500x500mm **See Photos No.1**
Test Type	Bullet Attack Resistance Test
Test Standard	Test Method - VPAM PM-2007 (Class 9) Triangle -Hit Test
Test Site	Ballistic Testing Facility (JODDB)
Samples Received Date	4th September 2022
Test Temperature	22 °C
Test Weapon	NATO Universal Ballistic Breech with standard barrel:7.62x51 mm SN:3710
Type of Bullets	308 Win "P 80" Full Steel Jacket (Plated), Pointed Bullet, hard Core, Bullet Mass (9,7 ± 0,2) gram Core Mass (4,0 ± 0,1)gram , Batch No.132- FNB -08
Test Equipment	• Doppler Radar SN.3503-13-270-011 (± 0.4 m/s) • Digital caliper SN. 112167 (± 0.05 mm) • Temp .Hum. Meter SN.95941255 (± 0.3 °C) • Distance laser Measuring Device (Lacia) SN. 1030969032 (±2 mm)
Test Conducted By	• Eng. Ali Alsardyah • Eng. Sammer Obidat • Eng. Mohannad Akilan • Senior Technician: Hasan Aldebs. • Technician: Muath Anagreh.
Test Date	11th September 2022

Note: The results contained in this report are only valid for detailed above and the report shall not be reproduced except in full without approval of the laboratory.

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2. Test Results

Test Sample	Number of Shots	Bullet Velocity (m/s) 1st	2nd	3rd
Sample No.1 Combined 14+6 mm Mild steel plates with 46mm air Gape Combined 14+6mm Mild steel plate 500x500mm	**Three Shots** 308 Win "P 80" Full Steel Jacket (Plated),Pointed Bullet, hard Core , Bullet Mass (9,7 ± 0,2) gram Core Mass (4,0 ± 0,1)gram Test Range: 10±0.5 m Impact Velocity (Doppler): 820±10 m/s Striking Distance: 120 ± 10mm	**820.8**	**825.5**	**817.1**
Status : (CP / PP) **CP** – Complete Penetration / **PP** - Partial Penetration		**PP**	**PP**	**PP**
Test Results		Partial Penetration **(See Photo No. 2)**		
Sample No.2 Combined 14+8 mm Mild steel plates with 46mm air Gape Combined 14+8 mm Mild steel plates 500x500mm	**Three Shots** 308 Win "P 80" Full Steel Jacket (Plated),Pointed Bullet, hard Core , Bullet Mass (9,7 ± 0,2) gram Core Mass (4,0 ± 0,1)gram Test Range: 10±0.5 m Impact Velocity (Doppler): 820±10 m/s Striking Distance: 120 ± 10mm	**825.4**	**824.1**	**824.6**
Status : (CP / PP) **CP** – Complete Penetration / **PP** - Partial Penetration		**PP**	**PP**	**PP**
Test Results		Partial Penetration **(See Photo No. 2)**		
Sample No.3 Combined 14mm Mild steel plate + 6.25 (Hardox 500) mm with 46mm air Gape Combined 14+6 mm Mild steel plates 500x500mm	**Three Shots** 308 Win "P 80" Full Steel Jacket (Plated),Pointed Bullet, hard Core , Bullet Mass (9,7 ± 0,2) gram Core Mass (4,0 ± 0,1)gram Test Range: 10±0.5 m Impact Velocity (Doppler): 820±10 m/s Striking Distance: 120 ± 10mm	**821.6**	**821.4**	**817.9**
Status : (CP / PP) **CP** – Complete Penetration / **PP** - Partial Penetration		**PP**	**PP**	**PP**
Test Results		Partial Penetration **(See Photo No. 2)**		

Test Criteria :

CP : Projectile or Projectile fragment completely penetrates the test specimen **or** Penetrated by the stuck projectile fragment on the rear surface **or** opening on backside of specimen with a light passage **or** penetration in specified penetration indicator (witness sheet).

PP: No Projectile or Projectile fragment completely penetrates the test specimen , No Penetrated by the stuck projectile fragment on the rear surface , No opening on backside of specimen with a light passage and No penetration in specified penetration indicator (witness sheet)

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RANGE EQUIPMENT & CONFIGURATION

THE GUN

The test samples were shot from the NATO Universal Ballistic Breech with the appropriate standard barrel and bullet type to give projectile stability.

VELOCITY MEASUREMENT

The projectile velocity was measured using Doppler Radar System at 2.5 m from the test sample attack face with uncertainty 0.4 m/s

DISTANCE MESURMENTS

Test range distance was measured by laser measuring device with uncertainty 2 mm, striking distances were marked by a digital calibre with uncertainty 0.05 mm .

SUPPORT FIXTURE

The test samples were supported by a fixture that permits its position and attitude to be readily adjusted so that it is perpendicular to the line of flight of the bullet at the point of impact.

AIMING

The aiming process has been achieved by using standard laser bullets.

SHOT PLACEMENT

All shots hit the test samples in the required pattern, with the required place.

SPLINTER COLLECTING BOX & PENETRATION INDICATOR

An aluminium sheet with a thickness of 0.5 mm (AlCuMg1, F 40) has to be used as the penetration indicator. It has to be fixed in a distance of 150 mm ± 5 mm behind the test sample.

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Photo No. 1: Test sample before firing

Sample No.1:

Combined 14+6 mm Mild steel plates with 46mm air Gape Combined 14+6mm Mild steel plate/ Size 500x500mm.

Front view

Isometric view

Sample No.2:

Combined 14+8 mm Mild steel plates with 46mm air Gape Combined 14+8 mm Mild steel plates/Size 500x500mm

Front view

Isometric view

Sample No.3:

Combined 14mm Mild steel plate + 6.25 (Hardox 500) mm with 46mm air Gape Combined 14+6 mm Mild steel plates/Size 500x500mm

Front view

Isometric view

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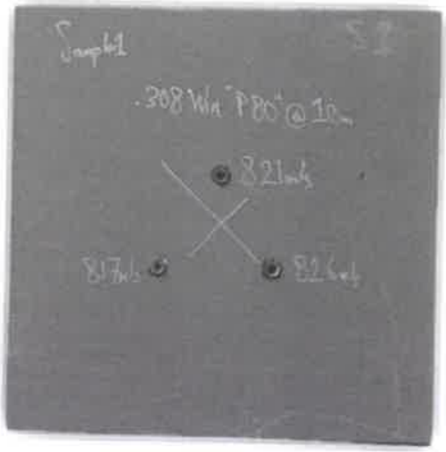
Photo No.2: Status of test samples after firing three rounds **308 Win "P 80" Full Steel Jacket (Plated),Pointed Bullet, hard Core , Bullet** Mass (9,7 ± 0,2) gram Core Mass (4,0 ± 0,1) gram on the impact side from 10 m.

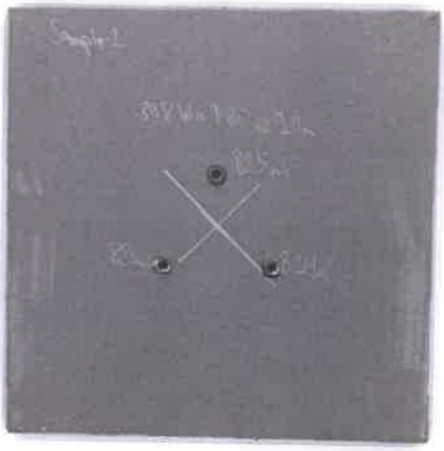
Sample No.1: **" Partial Penetration "**

Combined 14+6 mm Mild steel plates with 46mm air Gape Combined 14+6mm Mild steel plate/ Size 500x500mm.



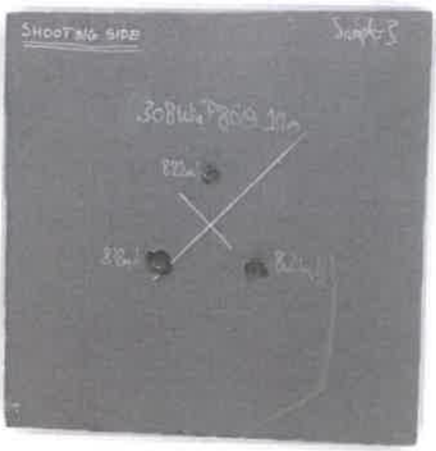
Sample No.2: **" Partial Penetration "**

Combined 14+8 mm Mild steel plates with 46mm air Gape Combined 14+8 mm Mild steel plates/Size 500x500mm



Sample No.3: **" Partial Penetration "**

Combined 14mm Mild steel plate + 6.25 (Hardox 500) mm with 46mm air Gape Combined 14+6 mm Mild steel plates/Size 500x500mm



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APPENDIX A: Approved Technical Testing Proposal

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Test and Evaluation Department
Ballistic Testing Facility
Technical Testing Proposal

Contractor	Special And Safety Work Nisreen Factory / KSA
Requester	JODDB/ Sales
Reference	Test and Evaluation Request No. 1086 dated 30/08/2022

Test Sample	Test Type	Test Reference	Test Requirements	Test Criteria
Combined 14+6 mm Mild steel plates with air Gape 46mm Combined 14+6mm Mild steel plates Dimension :500x500mm QTY: 1	Bullet Attack Resistance Test	Test Method VPAM PM-2007 (Class 9) "Triangle -Hit Test"	Three Strikes / each sample 308 Win "P 80" Full Steel Jacket (Plated), Pointed Bullet, hard Core , Bullet Mass (9,7 ± 0,2) gram Core Mass (4,0 ± 0.1)gram Shot Distance: 10 + 0.5 m Bullet Velocity: 820 ±10 m/s (Doppler Radar) Hit Distance :120±10 mm	See Annex A
Combined 14+8 mm Mild steel plates with air Gape 46mm Combined 14+8 mm Mild steel plates Dimension :500x500mm QTY: 1				
Combined 14mm Mild steel plate + 6.25 (Hardox 500) mm with air Gape 46mm Combined 14+6 mm Mild steel plates Dimension :500x500mm QTY: 1				

Note No. 1: It is necessary to provide JODDB with Steel heat numbers.

Note No. 2: The output will be Summary Test Results.

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Annex A

Test Criteria Statement of Conformity and Decision Rule			
Test Level	**Comply**	**Comply with Conditions**	**Not Comply**
VPAM PM 2007-Class 9 "Triangle Shooting" 308 Win "P 80"	No Projectile or Projectile fragment completely penetrates the test specimen AND No Penetrated by the stuck projectile fragment on the rear surface AND No opening on backside of specimen with a light passage AND No penetration in specified penetration indicator (witness sheet) When Bullet velocity ≥ 811.4 m/s	-	Projectile or Projectile fragment completely penetrates the test specimen or Penetrated by the stuck projectile fragment on the rear surface or opening on backside of specimen with a light passage or penetration in specified penetration indicator (witness sheet) When Bullet velocity ≤ 828.6 m/s

END OF THE SUMMARY TEST RESULTS

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Contractor	
Reference	
Contact Information	

According to ISO 17025:2017 requirements and in order to improve our services, Appreciate your cooperation by filling the following survey at your convenient time

QUALITY ELEMENT	Excellent	Good	Average	Fair	Poor
Technical Competence					
Turn-around Testing Time					
Communication with laboratory's employees					
Overall Quality Rating					
Compared to your experience with other laboratory providers, JODDB Testing Facilities services are					

Additional Comments:

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____

____

Name:

Date:

Job Title/Function: